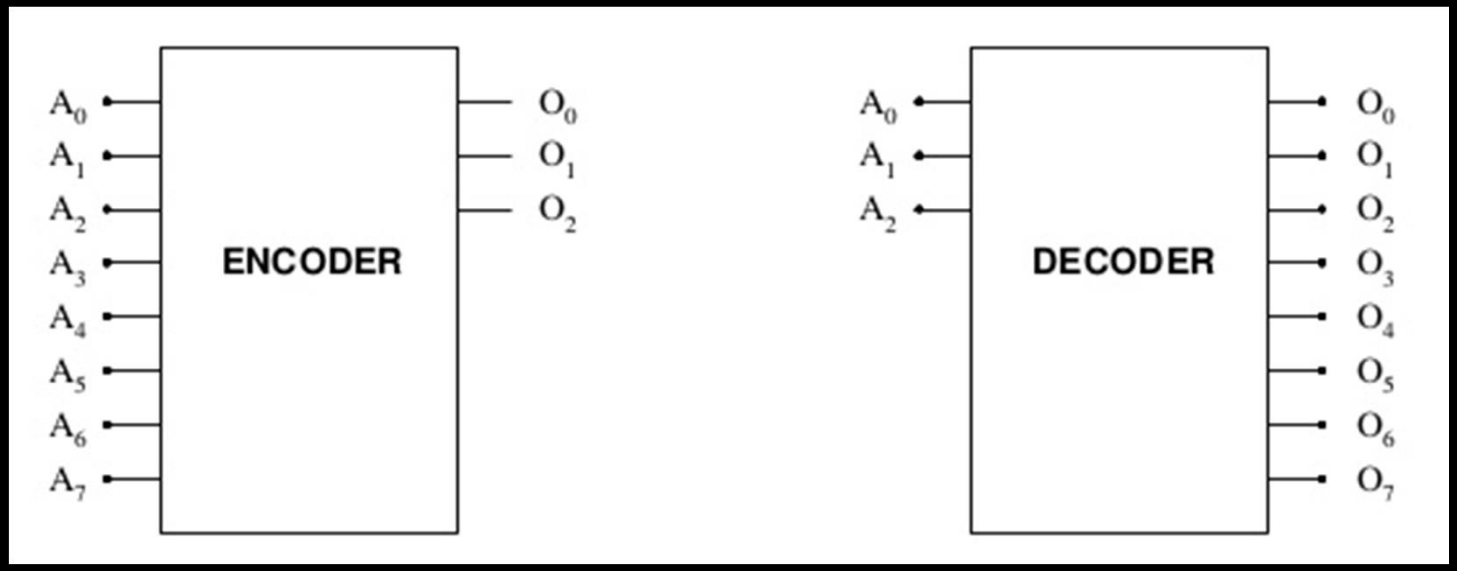
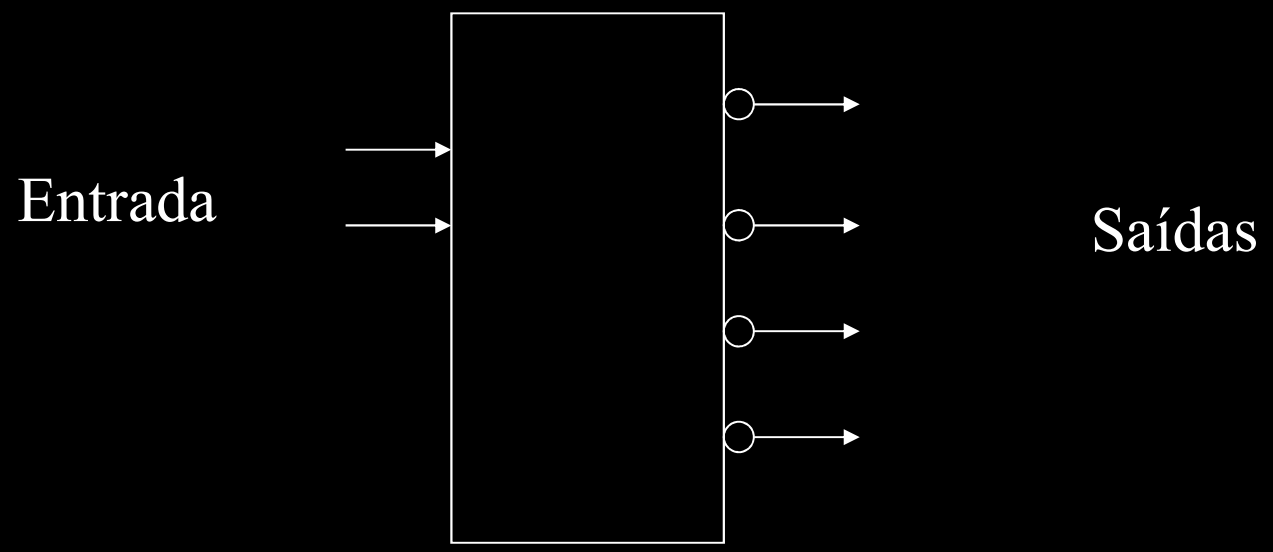
Circuitos Digitais I

Circuitos Codificadores e Decodificadores



Circuitos Decodificadores

Recebe um conjunto de entradas e ativa um conjunto de saídas a partir do especificado



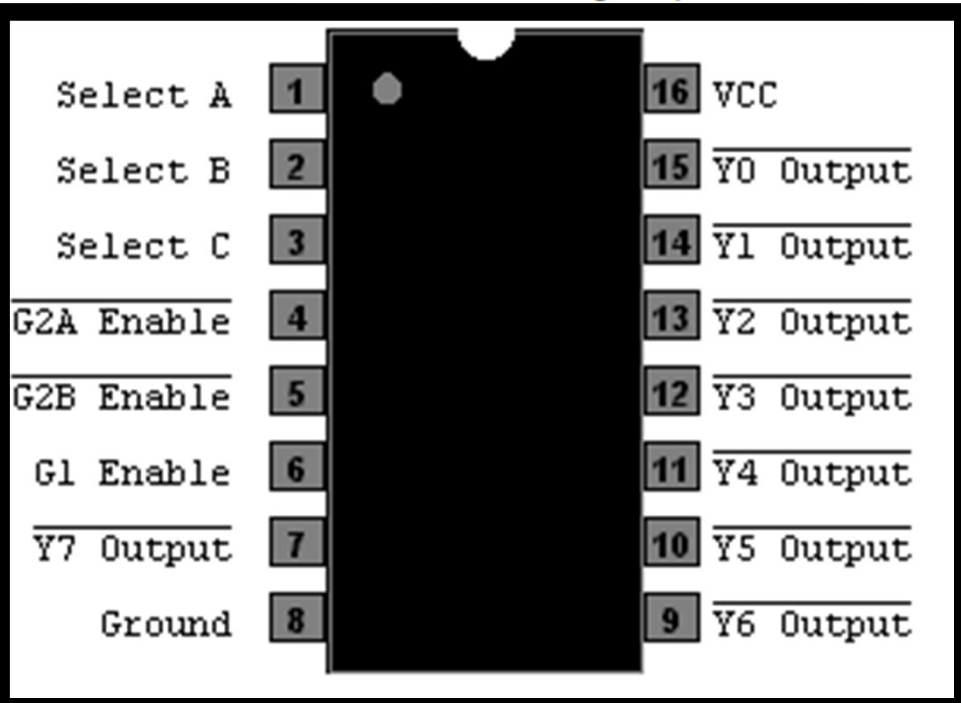
Circuitos Digitais I

Circuitos Decodificadores

Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/



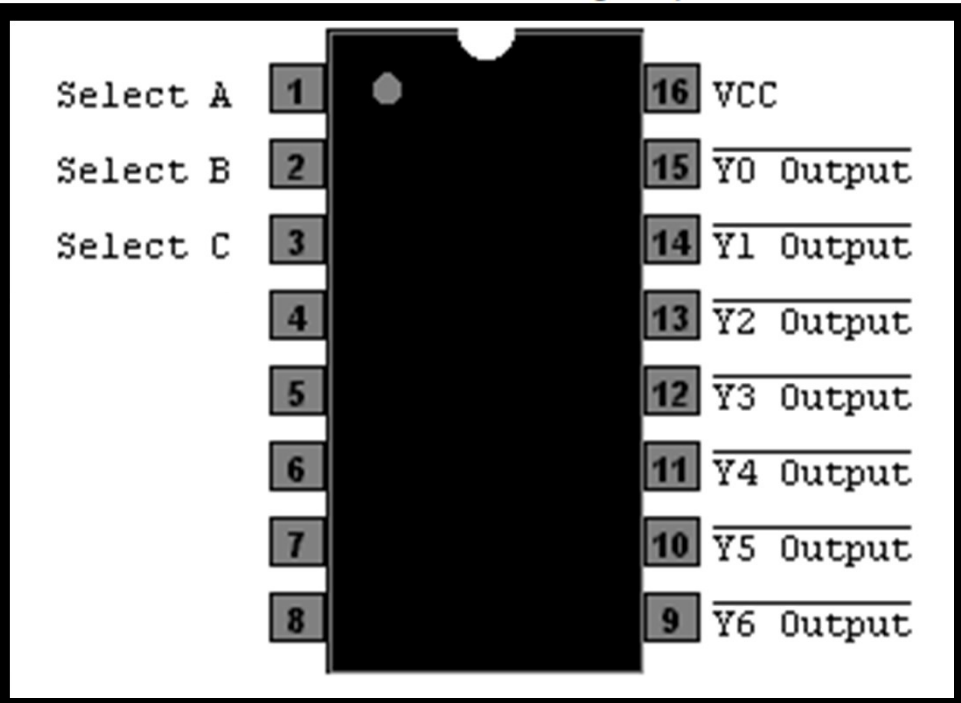
Circuitos Digitais I

Circuitos Decodificadores

Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/



Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input						Output							
$\overline{E1}$	$\overline{E2}$	E3	A0	A1	A2	$\overline{Y0}$	$\overline{Y1}$	$\overline{Y2}$	$\overline{Y3}$	$\overline{Y4}$	$\overline{Y5}$	$\overline{Y6}$	$\overline{Y7}$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

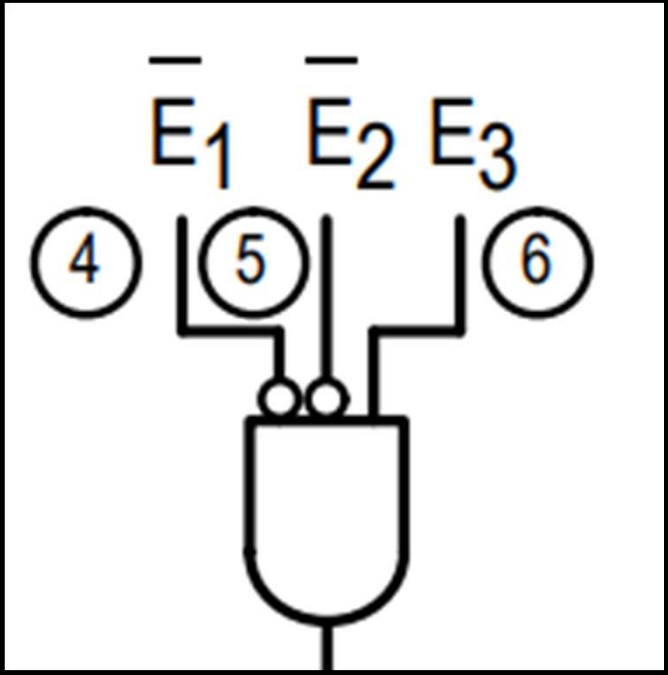
Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input		
$\overline{E1}$	$\overline{E2}$	E3
H	X	X
X	H	X
X	X	L
L	L	H
L	L	H
L	L	H
L	L	H
L	L	H
L	L	H
L	L	H
L	L	H

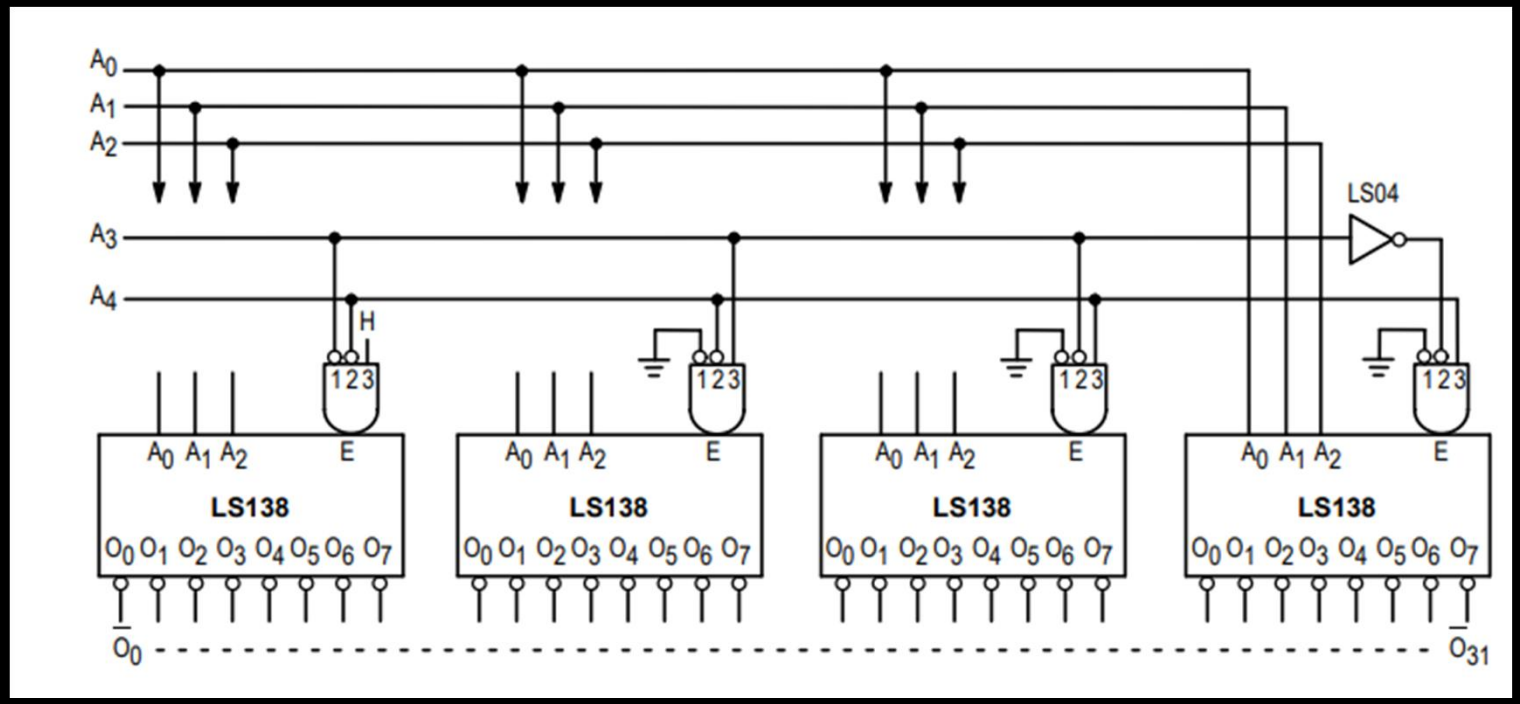


Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/



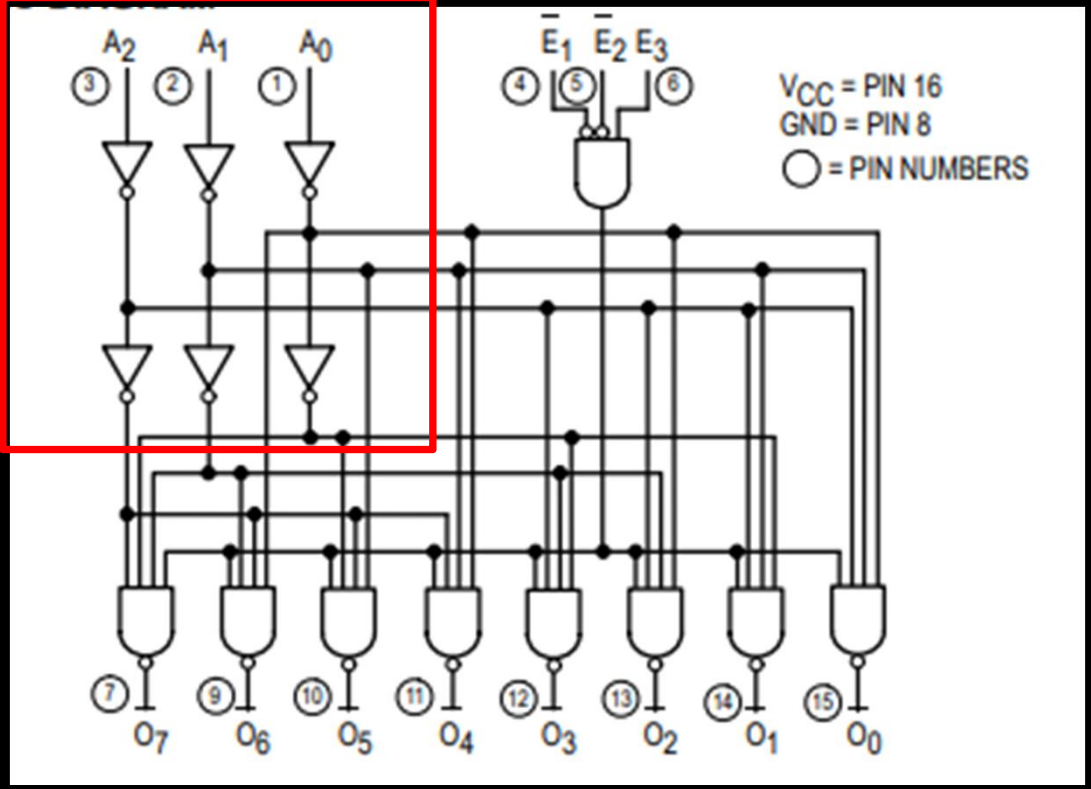
Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

A0	A1	A2
X	X	X
X	X	X
X	X	X
L	L	L
H	L	L
L	H	L
H	H	L
L	L	H
H	L	H
L	H	H
H	H	H



Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input						Output							
$\overline{E1}$	$\overline{E2}$	E3	A0	A1	A2	$\overline{Y0}$	$\overline{Y1}$	$\overline{Y2}$	$\overline{Y3}$	$\overline{Y4}$	$\overline{Y5}$	$\overline{Y6}$	$\overline{Y7}$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input						Output							
$\overline{E1}$	$\overline{E2}$	E3	A0	A1	A2	$\overline{Y0}$	$\overline{Y1}$	$\overline{Y2}$	$\overline{Y3}$	$\overline{Y4}$	$\overline{Y5}$	$\overline{Y6}$	$\overline{Y7}$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input						Output							
$\overline{E1}$	$\overline{E2}$	E3	A0	A1	A2	$\overline{Y0}$	$\overline{Y1}$	$\overline{Y2}$	$\overline{Y3}$	$\overline{Y4}$	$\overline{Y5}$	$\overline{Y6}$	$\overline{Y7}$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

Circuitos Digitais I

Circuitos Decodificadores: Alguns exemplos

1-OF-8 DECODER/ DEMULTIPLEXER

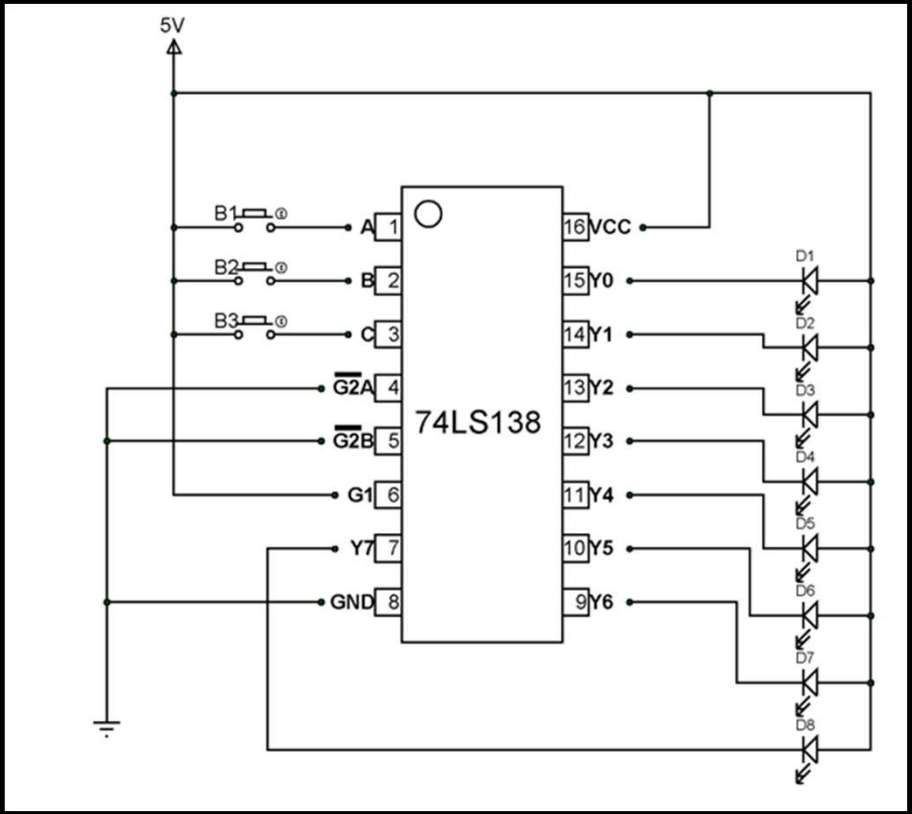
The LSTTL/MSI SN54/74LS138 is a high speed 1-of-8 Decoder/

Input						Output							
$\overline{E1}$	$\overline{E2}$	E3	A0	A1	A2	$\overline{Y0}$	$\overline{Y1}$	$\overline{Y2}$	$\overline{Y3}$	$\overline{Y4}$	$\overline{Y5}$	$\overline{Y6}$	$\overline{Y7}$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

Circuitos Digitais I

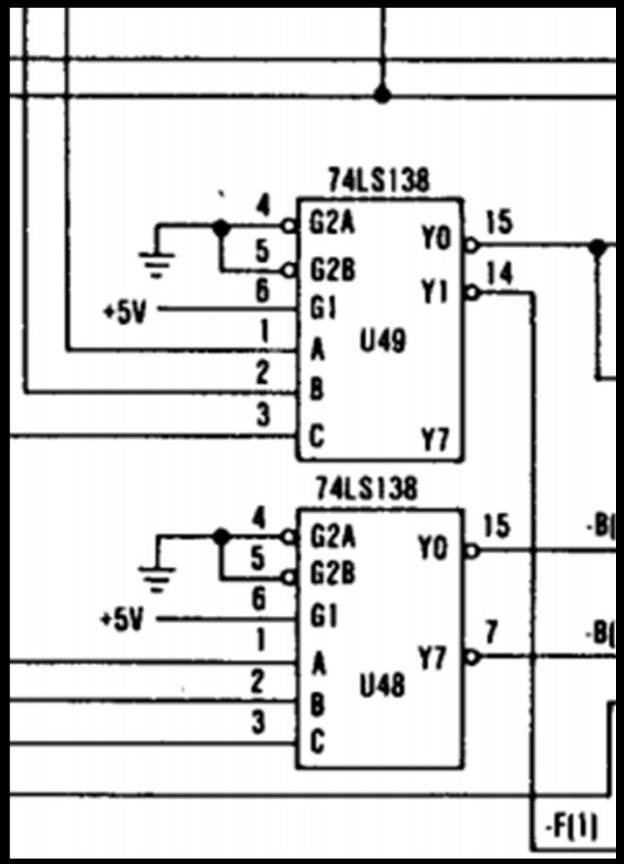
Circuitos Decodificadores

Alguns exemplos

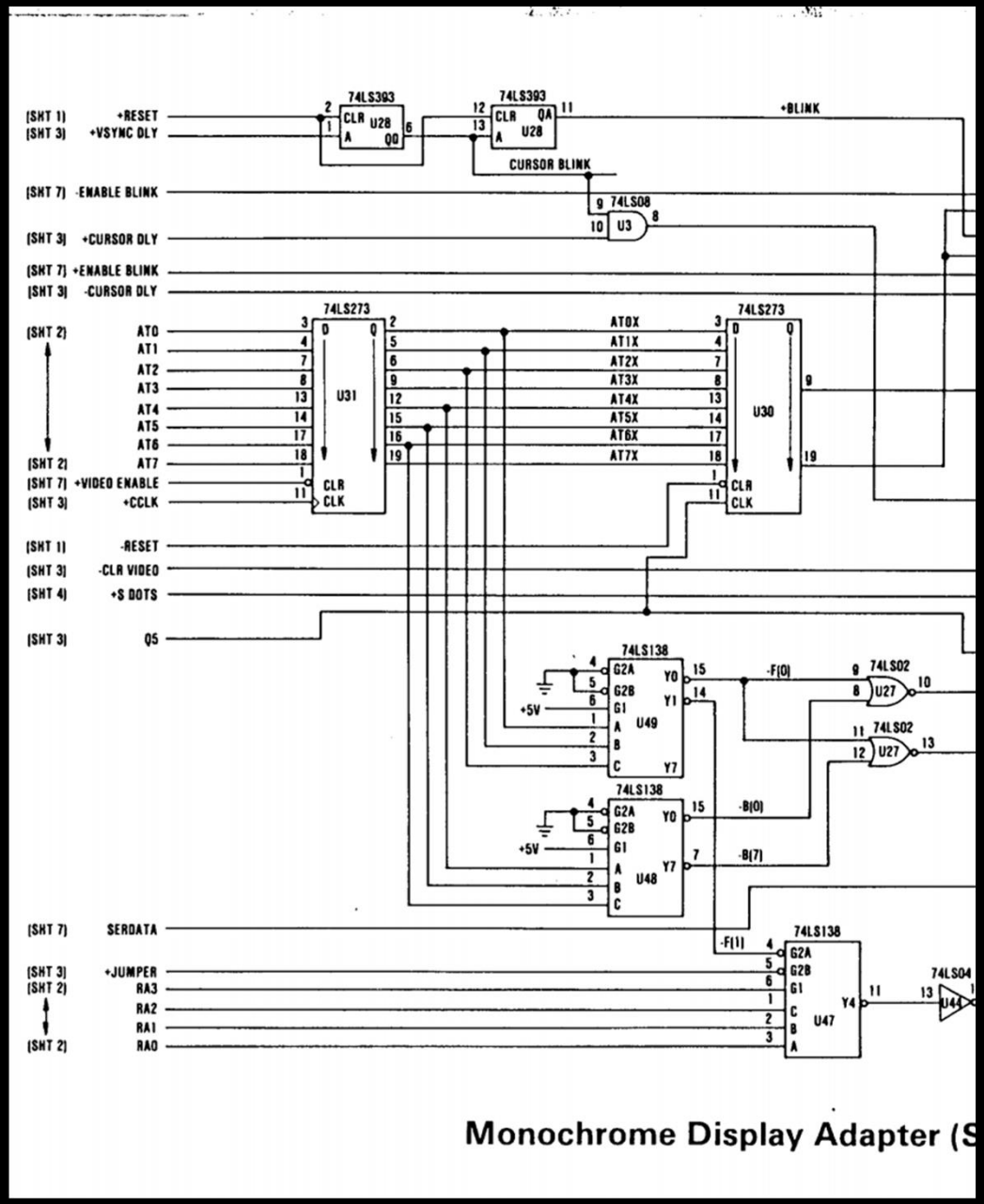


Circuitos Digitais I
Circuitos Decodificadores

Alguns exemplos



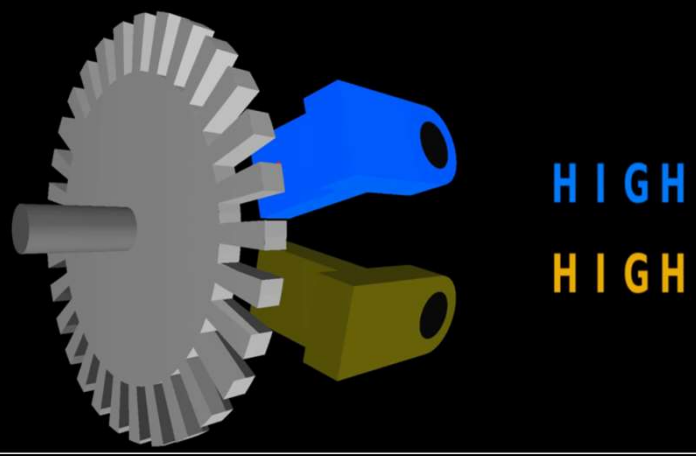
Detalhe IBM PC XT

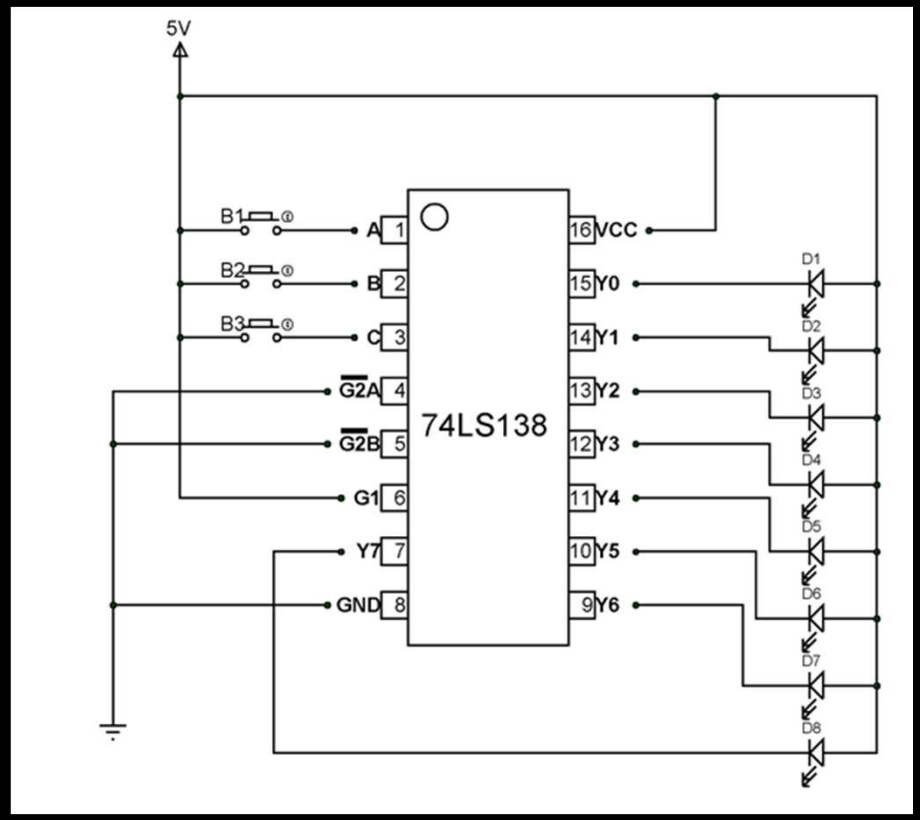


Circuitos Digitais I

Circuitos Decodificadores

Alguns exemplos

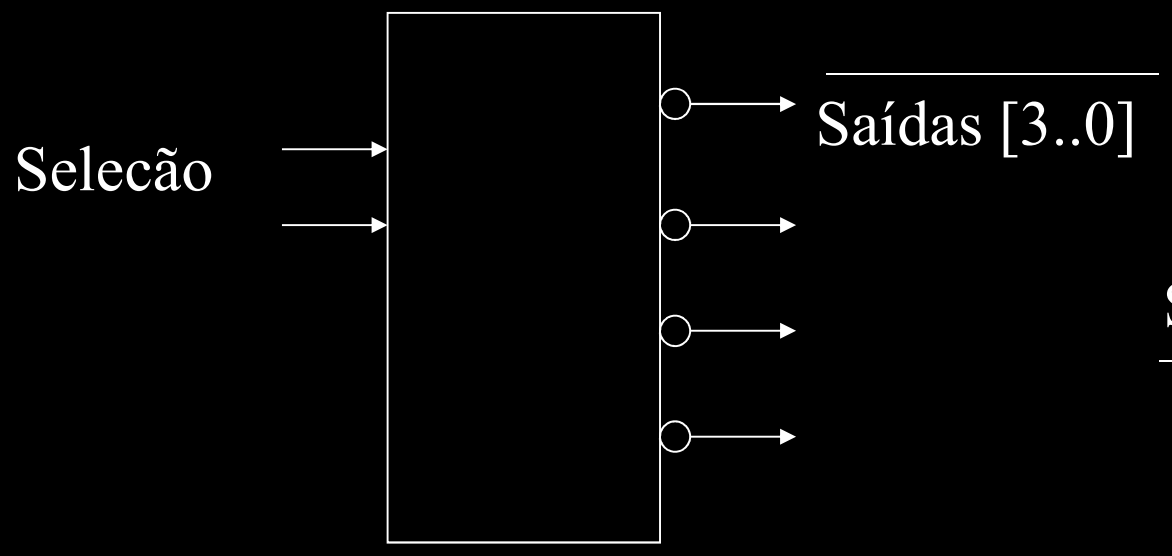




Circuitos Digitais I

Circuitos decodificadores

Um decodificador utiliza um valor binário ou endereço para selecionar um número de saídas



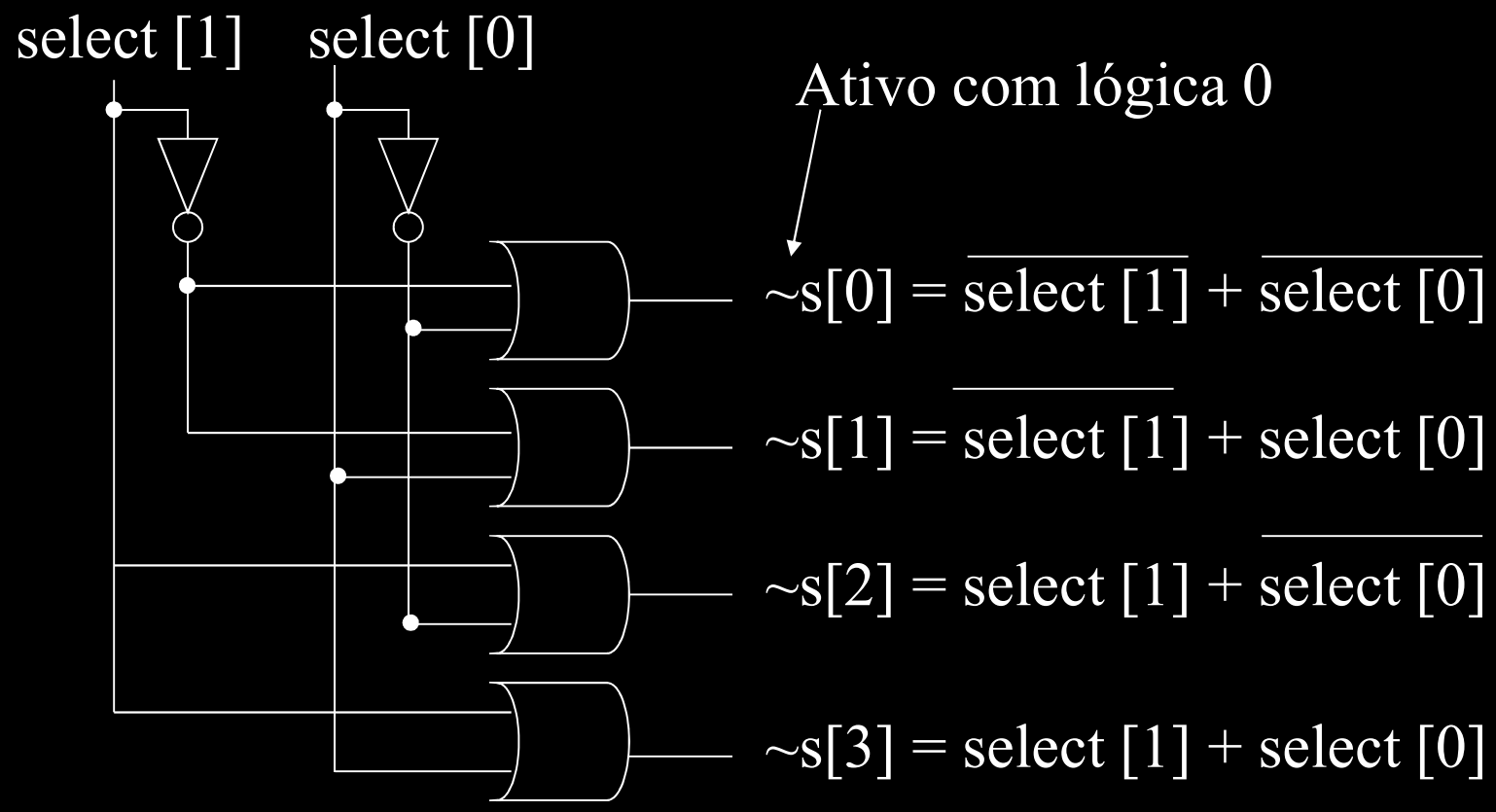
Selecao	Saídas [3..0]
0 0	1 1 1 0
0 1	1 1 0 1
1 0	1 0 1 1
1 1	0 1 1 1

Decodificador 2 para 4

Circuitos Digitais I

Circuitos decodificadores

Um decodificador utiliza um valor binário ou endereço para selecionar um número de saídas



Circuitos Digitais I

Circuitos decodificadores Codificador de prioridade 8-3

Aplicação: Detectar e tratar Interrupções em CPU

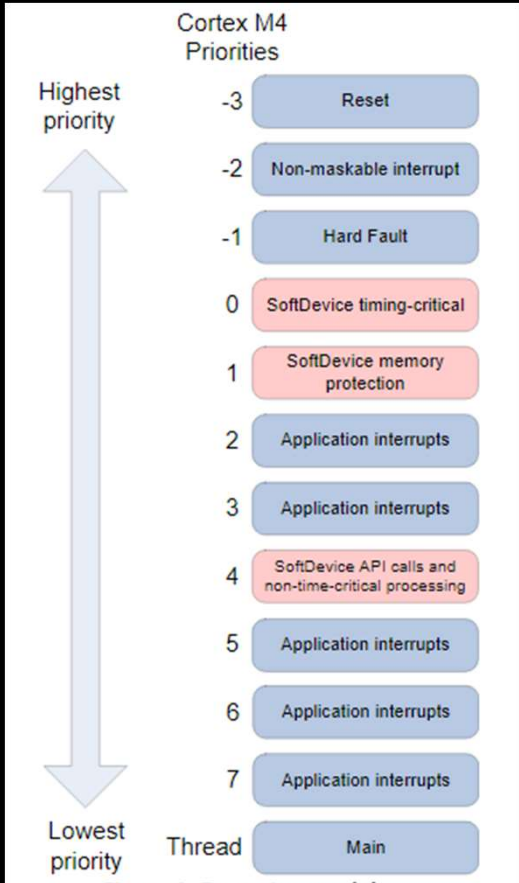
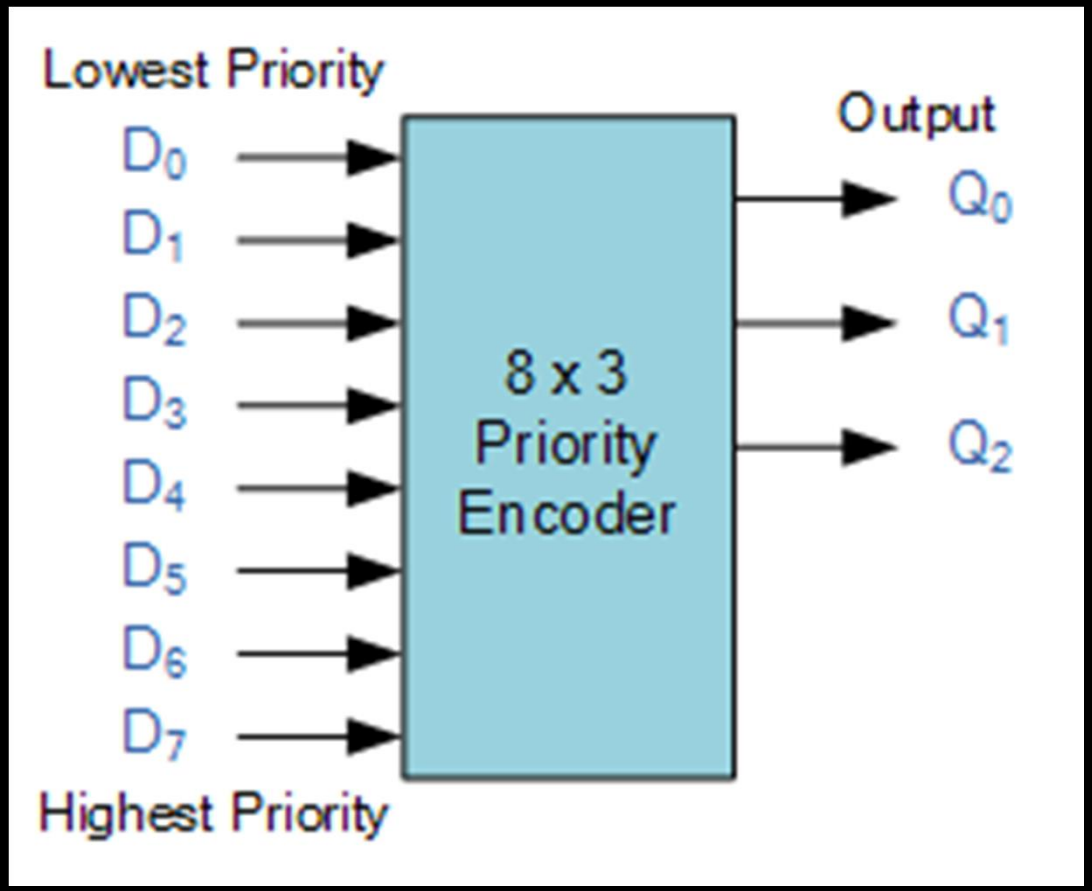


Figure 1. Exception model

Circuitos Digitais I

Circuitos decodificadores

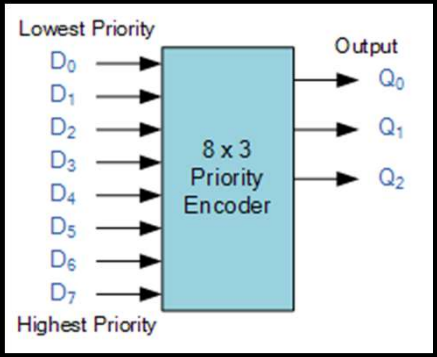
Codificador de prioridade 8-3



Circuitos Digitais I

Circuitos decodificadores

Codificador de prioridade 8-3



Digital Inputs								Binary Output		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	X	0	0	1
0	0	0	0	0	1	X	X	0	1	0
0	0	0	0	1	X	X	X	0	1	1
0	0	0	1	X	X	X	X	1	0	0
0	0	1	X	X	X	X	X	1	0	1
0	1	X	X	X	X	X	X	1	1	0
1	X	X	X	X	X	X	X	1	1	1

Circuitos Digitais I

Circuitos decodificadores

Codificador de prioridade 8-3

Digital Inputs								Binary Output		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	X	0	0	1
0	0	0	0	0	1	X	X	0	1	0
0	0	0	0	1	X	X	X	0	1	1
0	0	0	1	X	X	X	X	1	0	0
0	0	1	X	X	X	X	X	1	0	1
0	1	X	X	X	X	X	X	1	1	0
1	X	X	X	X	X	X	X	1	1	1

$$Q_0 = \overline{D}_7\overline{D}_6\overline{D}_5\overline{D}_4\overline{D}_3\overline{D}_2D_1 + \overline{D}_7\overline{D}_6\overline{D}_5\overline{D}_4D_3 + \overline{D}_7\overline{D}_6D_5 + D_7$$

$$Q_0 = \overline{D}_6\left(\overline{D}_4\overline{D}_2D_1 + \overline{D}_4D_3 + D_5\right) + D_7$$

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário

BCD (Binary Coded Decimal)

- Forma de expressar os dígitos decimais na forma binária
- Representa os dígitos de 0 a 9
- Valores inválidos 10 a 15
- Formato 8421

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário

BCD (Binary Coded Decimal) - Exemplos

- 0010 => 2
- 0011 => 3
- 1001 => 9
- 1101 => Inválido

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário
ASCII (American Standard Code for Information Interchange)

- 128 caracteres e símbolos para representar caracteres e símbolos
- representado em código binário de 7 bits
- 32 primeiros utilizados para propósito de controle
- conjunto estendido de 128 caracteres para caracteres estrangeiros

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário
ASCII (American Standard Code for Information Interchange)

- 128 caracteres e símbolos para representar caracteres e símbolos
- representado em código binário de 7 bits
- 32 primeiros utilizados para propósito de controle
- conjunto estendido de 128 caracteres para caracteres estrangeiros

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário: ASCII

	coluna								
	bits	0	1	2	3	4	5	6	7
linha	7654321	000	001	010	011	100	101	110	111
0	0000	NUL	DLE	SP	0	@	P	`	p
1	0001	SOH	DC1	!	1	A	Q	a	q
2	0010	STX	DC2	"	2	B	R	b	r
3	0011	ETX	DC3	#	3	C	S	c	s
4	0100	EOT	DC4	$	4	D	T	d	t
5	0101	ENQ	NAK	%	5	E	U	e	u
6	0110	ACK	SYN	&	6	F	V	f	v
7	0111	BEL	ETB	'	7	G	W	g	w
8	1000	BS	CAN	(	8	H	X	h	x
9	1001	HT	EM	)	9	I	Y	i	y
10	1010	LF	SUB	*	:	J	Z	j	z
11	1011	VT	ESC	+	;	K	[	k	{
12	1100	FF	FS	,	<	L	\	l	|
13	1101	CR	GS	-	=	M	]	m	}
14	1110	SO	RS	.	>	N	^	n	~
15	1111	SI	US	/	?	O	_	o	DEL

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário: ASCII

MSB

LSB

		coluna					
	bits	0	1	2	3	4	5
linha	7654321	000	001	010	011	100	101
0	0000	NUL	DLE	SP	0	@	P
1	0001	SOH	DC1	!	1	A	Q
2	0010	STX	DC2	"	2	B	R
3	0011	ETX	DC3	#	3	C	S

Código do Caractere = 3 bits | 4 bits

Circuitos Digitais I

Circuitos decodificadores

Códigos Binário: ASCII

MSB

LSB

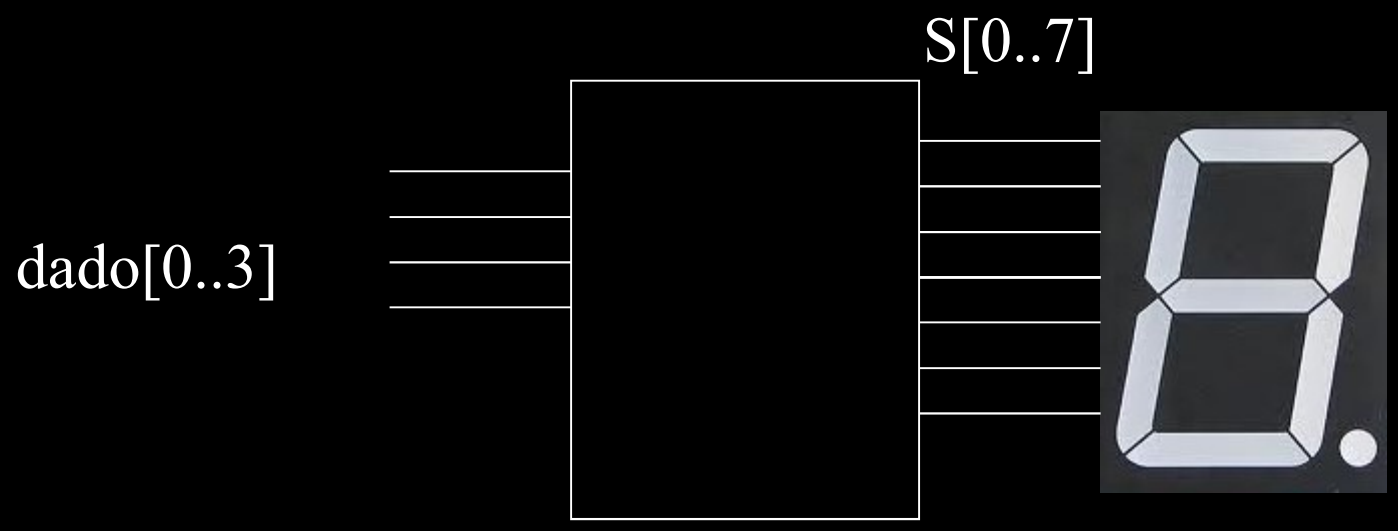
	coluna						
	bits	0	1	2	3	4	5
linha	7654321	000	001	010	011	100	101
0	0000	NUL	DLE	SP	0	@	P
1	0001	SOH	DC1	!	1	A	Q
2	0010	STX	DC2	"	2	B	R
3	0011	ETX	DC3	#	3	C	S

Código do Caractere A = 100 0001

Circuitos Digitais I

Circuitos decodificadores

Decodificador display 7 segmentos

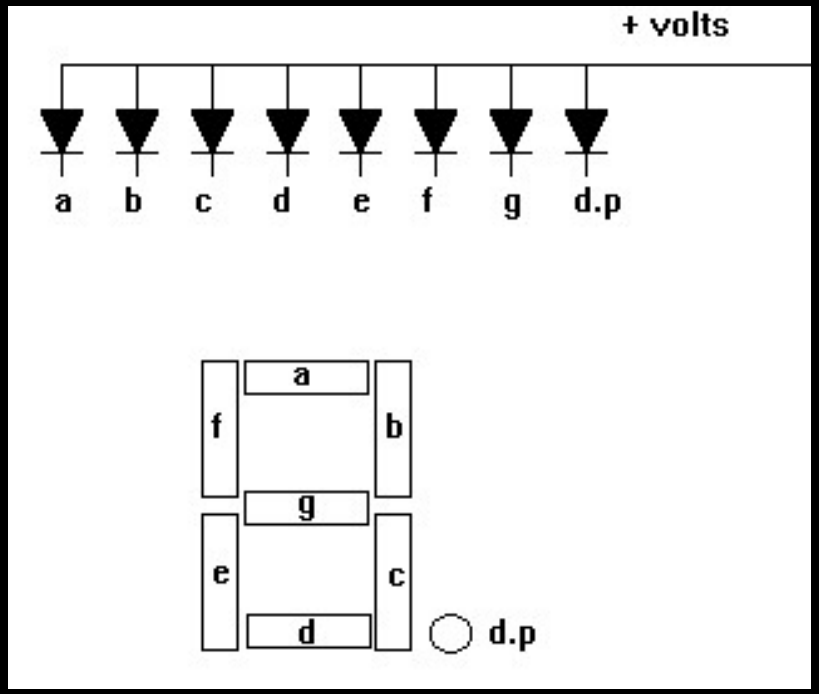


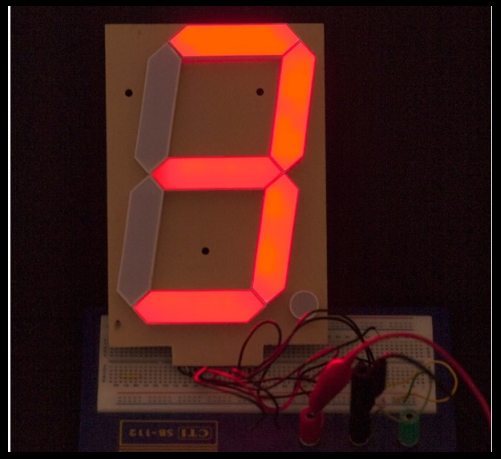
Circuitos Digitais I

Circuitos decodificadores

Decodificador display 7 segmentos

Conjunto de LED´s com anodo comum



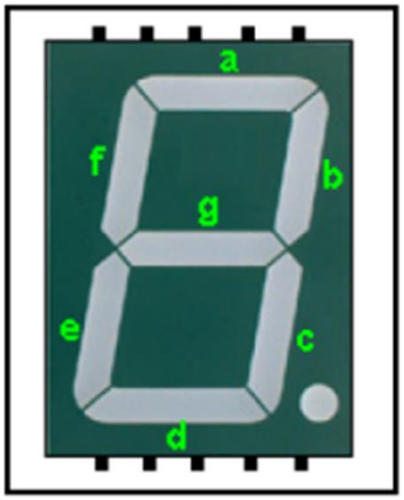


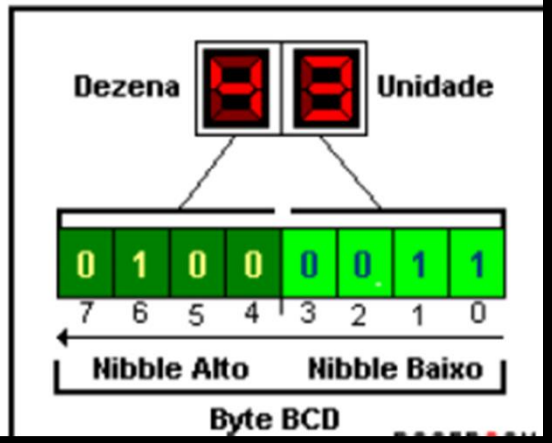
Circuitos Digitais I

Decodificador display 7 segmentos

Tabela 1 - Informações e codificação do display

entradas BCD				segmentos de saída							DISPLAY
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0	1	2
0	0	1	1	1	1	1	1	0	0	1	3
0	1	0	0	0	1	1	0	0	1	1	4
0	1	0	1	1	0	1	1	0	1	1	5
0	1	1	0	1	0	1	1	1	1	1	6
0	1	1	1	1	1	1	0	0	0	0	7
1	0	0	0	1	1	1	1	1	1	1	8
1	0	0	1	1	1	1	1	0	1	1	9





Circuitos Digitais I

Decodificador display 7 segmentos

Decimal	BCD 8421				7 segmentos						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
10	x	x	x	x	x	x	x	x	x	x	x
11	x	x	x	x	x	x	x	x	x	x	x
12	x	x	x	x	x	x	x	x	x	x	x
13	x	x	x	x	x	x	x	x	x	x	x
14	x	x	x	x	x	x	x	x	x	x	x
15	x	x	x	x	x	x	x	x	x	x	x

Circuitos Digitais I

Decodificador display 7 segmentos

$$(a) = A + C + BD + \overline{B}\,\overline{D} = A + C + \overline{B \oplus D}$$

$$(b) = \overline{B} + \overline{C}\,\overline{D} + CD = \overline{B} + \overline{C \oplus D}$$

$$(c) = B + \overline{C} + D$$

$$(d) = A + \overline{B}\,\overline{D} + \overline{B}C + C\overline{D} + B\overline{C}D$$

$$(e) = \overline{BD} + C\overline{D}$$

$$(f) = A + \overline{C}\,\overline{D} + B\overline{C} + B\overline{D}$$

$$(g) = A + B\overline{C} + \overline{B}C + C\overline{D}$$